

BMGT 240IS — Business Analytics

Fall 2026 • Section 002 • CRN 23266

MEETS Tue & Thu, 12:15–1:30 pm · JABS 215	TERM Aug 27 – Dec 17, 2026 · 3 credits	INSTRUCTOR Xuying W. Swift
EMAIL xuying.swift@montana.edu	OFFICE HOURS <i>TBD — posted on Canvas in week 1</i>	PLATFORMS Canvas · Cengage MindTap

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Course Description

This course enables students to understand basic statistical concepts and techniques and how modern managers apply them in business. Emphasis is on statistical reasoning, problem-solving, computer applications, and interpretation of results.

The through-line is one question asked in progressively harder forms: **what can I responsibly conclude from imperfect data?** We describe data, then reason from a sample to a population, then test claims, then model relationships among several variables.

Learning Objectives

#	YOU WILL BE ABLE TO...	ASSESSED BY
1	Gather, clean, describe, and summarize data using Excel	MindTap Ch 1–3 · Project 1 · Midterm
2	Take samples and estimate unknown parameters	MindTap §6.2, Ch 7–8 · Project 2 · Midterm
3	Formulate and test hypotheses	MindTap Ch 9–13 · Project 2 · Final Project
4	Understand basic principles of statistical process control	<i>see note</i>
5	Use multiple regression to analyze data	MindTap Ch 14–15 · Final Project report and presentation
6	Successfully complete research projects	Projects 1 and 2 · Final Project
7	Design and conduct research in the future	Final Project proposal, report, and presentation

- | # | YOU WILL BE ABLE TO... | ASSESSED BY |
|---|---|---|
| 8 | Use sound quantitative reasoning to analyze business problems | Interpretation sections of all projects and exams |

Note on objective 4. The *Essentials* edition has no statistical process control chapter — SPC appears only in the unabridged *Modern Business Statistics*. No session is allocated to it. If it is assessed this term, a short introduction will be posted to Canvas and announced in class; otherwise objectives 1–3 and 5–8 are the graded set.

Materials

Cengage MindTap — required. All quizzes and chapter assignments are completed in MindTap, linked from Canvas. Your MindTap access includes the **eBook**, so no separate textbook purchase is necessary. Get access in week 1; missed deadlines caused by a late purchase are not excused.

Textbook. Camm, Cochran, Fry, Ohlmann, Anderson, Sweeney & Williams, *Essentials of Modern Business Statistics with Microsoft Excel*, 9th edition (Cengage, 2025). ISBN 978-0-357-93004-5. A print copy is optional. **This is a new textbook this year** — chapter numbering does not match older editions or older notes. Use the section numbers in the schedule below.

Technology

- **Microsoft Excel, desktop version**, with the **Analysis ToolPak** enabled. Free through MSU's Microsoft 365. The browser version will *not* run tools we depend on. We set this up together in session 2.
- A **laptop** in class on days marked **Lab**. If a laptop is a problem, tell me in week 1 and we will solve it quietly.
- A **backup** of your work. A lost workbook is not an extension.

How the course runs

Tuesdays introduce a method — where it comes from, what it assumes, what it is for. Thursdays continue it and put it to work in Excel. **MindTap** work for each chapter is due at **11:59 pm MT** on the date shown in the schedule.

Projects are due at the start of class on the date shown, submitted on Canvas.

We do not cover all fifteen chapters. See the note after the schedule for what is omitted and why.

Grading

COMPONENT	WEIGHT	POINTS
MindTap quizzes — 11 given, lowest dropped (10 × 15)	15%	150
MindTap chapter assignments — best 10 of 11 (10 × 5)	5%	50
Project 1	10%	100
Project 2	10%	100
Midterm Exam — Thu, Oct 15	25%	250
Final Project — team; proposal, report, and presentation	35%	350
Total	100%	1000

GRADE	POINTS	GRADE	POINTS
A	930–1000	C	730–769
A–	900–929	C–	700–729
B+	870–899	D+	670–699
B	830–869	D	640–669
B–	800–829	D–	600–639
C+	770–799	F	below 600

Points are what count. Final scores are **not** rounded up across a boundary — 929 points is an A–.

There is no final exam. The midterm on Oct 15 is the only exam in this course. Your final exam period is used for **Final Project presentations** instead — attendance at it is required, and it is graded.

Extra credit. Opportunities may be announced during the semester — the Jabs Career Symposium is the usual one — for up to **15 points**. Nothing is guaranteed and none is offered individually on request.

Midterm Exam · 250 points

	WHEN	COVERS
Midterm	Thu, Oct 15, in class	Ch 1–3, §6.2, 7–8

The midterm closes the describe-and-estimate half of the course. It is the only exam, so it carries a quarter of your grade on its own — treat the date as fixed.

It is closed-book. A **formula sheet is provided** and you may bring **one double-sided page of handwritten notes**. Bring your laptop; Excel is used for computation.

Make-ups are given only for documented, university-sanctioned conflicts arranged **at least a week in advance**. Work travel and vacations are not excuses. With no second exam to absorb it, a missed midterm is not recoverable.

Projects · 550 points

Written work has two submission steps: upload your **workbook and written report** to Canvas › Assignments, then complete the **project quiz** that opens conditional on that upload. Missing or incorrect files reduce or zero the project quiz. Emailed work counts as not submitted. **Late work is not accepted** — the window closes at the start of class.

Project 1 and Project 2 · 100 points each · individual work

	POINTS
Correct analysis — right method, right execution	40
Written interpretation — what it means, what it does not mean, what you would do	45
A workbook someone else could follow — labeled sheets, live formulas, documented sources	15

- **Project 1 — Describing a business dataset.** Assigned Sep 17, due **Tue, Oct 6**. Ch 1–3: choose and defend displays, compute and interpret location, variability, and association.
- **Project 2 — Estimating and testing.** Assigned Oct 20, due **Tue, Nov 10**. Ch 7–9: build a confidence interval and run a hypothesis test, state your assumptions, and write the conclusion in business language.

Instructions and data are posted to Canvas on the day each is assigned. Both are individual work and are checked for plagiarism.

Final Project · 350 points · teams of four

Real small-scale research on a business question you choose. Pose the question, obtain data, analyze it — **multiple regression must be part of the analysis** — report what you found in writing, and present it to the class. Teams are formed in class **Thu, Oct 22**.

MILESTONE	DUE	POINTS
Topic proposal — one page: the question, the data, the methods you expect to need	Tue, Nov 17	25
Written report + workbook	Tue, Dec 8	225
Presentation	Dec 10 or finals week	100

Report scoring, out of 225: question and data (30) · descriptive and graphical analysis (35) · inference, with assumptions stated (35) · **multiple regression — specification, output, interpretation, diagnostics (60)** · written report, including honest limitations (50) · reproducible workbook (15).

The proposal is worth 25 points but it is the most useful thing you will submit. Most failed student research projects fail at the question, not the statistics — a question no obtainable dataset can answer, or one whose answer is already obvious. The proposal exists to catch that three weeks early.

Peer evaluation. Every member submits a confidential evaluation of teammates' contributions, which can adjust an individual score by up to $\pm 10\%$. A member who does not contribute does not receive the team's grade.

FINAL PROJECT PRESENTATION · 100 POINTS

12 minutes, plus 3 minutes of questions. Every member speaks. Slides are uploaded to Canvas by 11:59 pm MT the night before your team presents; you present from the posted file, so nothing is edited between upload and delivery.

Teams present in one of two rounds — **Thu, Dec 10** in class, or during the **final exam period, Dec 14–17**. Rounds are assigned by random draw announced Dec 1. Both rounds are required attendance for everyone: you are an audience member on the day you are not presenting, and audience participation counts toward the delivery score.

	POINTS
The question — what you asked and why a decision-maker should care	15
Method — what you did and why it was the right choice for this question	25
Findings — your results, including the regression, read correctly	30
Limitations — what your data and model cannot tell you, stated plainly	15
Delivery — timing, legible slides, every member contributing, questions handled	15

Because the report is due Dec 8, your analysis is finished before anyone presents. The presentation is not new work; it is the argument you already wrote, made in twelve minutes to people who have not read it. That translation is the skill being graded.

A presentation is not a reading of your report. The most common failure is a team narrating every table in order and running out of time before the conclusion. Lead with what you found. Then show only the evidence that supports it.

MindTap Quizzes and Assignments · 200 points

Each covered chapter has one **quiz** (15 pts) and one **chapter assignment** (5 pts) in MindTap, due at **11:59 pm MT** on the date in the schedule. Eleven of each are given; the **lowest quiz is dropped** and only your **best ten assignments** count.

Those drops are the make-up policy. They cover illness, travel, a bad week, and a forgotten date, so **there are no make-ups and no individually excused deadlines**. MindTap deadlines are enforced by the platform, not by me. If you will miss more than one week for a documented reason, talk to me in advance.

This is your own individual work.

Course Schedule

Twenty-nine meetings following Camm et al., 9e. Section numbers refer to that edition. Tentative — changes are announced in class, posted to Canvas, and emailed.

— HANDS-ON LAB — SOMETHING DUE — EXAM — NO CLASS

UNIT 1 Describing Data

Ch 1–3 · Aug 27 – Sep 17

#	DATE	TOPIC	READ	DUE
1	Thu, Aug 27	Course overview; Data and Statistics — data and variables, scales of measurement, categorical vs. quantitative, cross-sectional vs. time series	§1.1–1.3	
2	Tue, Sep 1	Descriptive statistics vs. inference; analytics, big data, ethical guidelines · Lab : Excel and the Analysis ToolPak	§1.4–1.9	
3	Thu, Sep 3	Tabular and Graphical Displays — frequency, relative, and percent frequency distributions; bar and pie charts	§2.1	MindTap Ch 1
4	Tue, Sep 8	Quantitative displays: class intervals, cumulative distributions, dot plots, histograms and skewness, stem-and-leaf · Lab	§2.2	
5	Thu, Sep 10	Two variables: crosstabulation with PivotTables, row and column percentages, Simpson's paradox, scatter diagrams, side-by-side and stacked bars, geospatial displays · Lab	§2.3–2.5	MindTap Ch 2
6	Tue, Sep 15	Numerical Measures — mean, median, mode, weighted and geometric mean, percentiles, quartiles	§3.1	
7	Thu, Sep 17	Variability and shape: range, IQR, variance, standard deviation, CV, z-scores, the empirical rule, outliers, boxplots; covariance and correlation · Lab : Descriptive Statistics tool	§3.2–3.5	MindTap Ch 3 · Project 1 assigned

UNIT 2 From Sample to Population

§6.2, Ch 7–8 · Sep 22 – Oct 15

#	DATE	TOPIC	READ	DUE
8	Tue, Sep 22	The normal distribution — the normal and standard normal curves, z, and probability as area · Lab : NORM.DIST, NORM.INV, NORM.S.INV	§6.2	
9	Thu, Sep 24	Sampling and Sampling Distributions — simple random sampling, point estimation, the sampling distribution of the sample mean, standard error, and the Central Limit Theorem · Lab	§7.1–7.5	
10	Tue, Sep 29	The sampling distribution of the sample proportion; stratified, cluster, systematic, and convenience sampling; why big data does not fix sampling error	§7.6–7.8	

#	DATE	TOPIC	READ	DUE
11	Thu, Oct 1	Interval Estimates — confidence interval for a mean, σ known and unknown; the t distribution; margin of error · Lab	§8.1–8.2	MindTap §6.2 + Ch 7
12	Tue, Oct 6	Determining the sample size; confidence interval for a proportion · Lab : CONFIDENCE.T, T.INV.2T	§8.3–8.4	Project 1 due
13	Thu, Oct 8	What a confidence interval does and does not claim; intervals when n is enormous; worked business cases · Lab	§8.5	MindTap Ch 8
14	Tue, Oct 13	Review for the Midterm	Ch 1–3, §6.2, 7–8	
15	Thu, Oct 15	MIDTERM EXAM — Ch 1–3, §6.2, 7–8		

UNIT 3 Hypothesis Testing

Ch 9 · Oct 20 – Oct 27

#	DATE	TOPIC	READ	DUE
16	Tue, Oct 20	Hypothesis Tests — developing null and alternative hypotheses; Type I and Type II errors; the logic of a test before any formula	§9.1–9.2	Project 2 assigned
17	Thu, Oct 22	One-tailed and two-tailed tests about a population mean, σ known and unknown; p -values · Lab	§9.3–9.4	Final Project assigned; teams formed
18	Tue, Oct 27	Tests about a population proportion; statistical vs. practical significance; what a p -value is not	§9.5–9.6	

UNIT 4 Comparing Two or More Populations

Ch 10, 12, 13 · Oct 29 – Nov 17

#	DATE	TOPIC	READ	DUE
19	Thu, Oct 29	Two Populations — inference about the difference between two means, σ 's known and unknown · Lab : Excel's t -Test and z -Test tools	§10.1–10.2	MindTap Ch 9
20	Tue, Nov 3	Matched samples; inference about the difference between two proportions	§10.3–10.4	
21	Thu, Nov 5	Categorical Data — the chi-square distribution and the goodness-of-fit test · Lab	§12.1	MindTap Ch 10
22	Tue, Nov 10	What independence means for two categorical variables; the test of independence; testing the equality of three or more proportions · Lab	§12.2–12.3	Project 2 due
23	Thu, Nov 12	Experimental Design and ANOVA — completely randomized designs, partitioning total variation, and the F test	§13.1–13.2	MindTap Ch 12
24	Tue, Nov 17	Multiple comparisons and Fisher's LSD; reading ANOVA output and reporting the conclusion · Lab : Anova: Single Factor	§13.3	Final Project proposal due

#	DATE	TOPIC	READ	DUE
25	Thu, Nov 19	Simple Linear Regression — the model, the least squares method, the coefficient of determination, model assumptions	§14.1–14.4	MindTap Ch 13
–	Nov 23–27	<i>Fall Break — no classes</i>		
26	Tue, Dec 1	Testing for significance; estimation and prediction; residual analysis; outliers and influential observations · Lab : Excel's Regression tool	§14.5–14.9	
27	Thu, Dec 3	Multiple Regression — least squares with several predictors, multiple and adjusted R^2 , the F test versus the individual t tests, interpreting a coefficient <i>holding the others constant</i> · Lab	§15.1–15.6	MindTap Ch 14
28	Tue, Dec 8	Categorical independent variables (dummy coding), residual analysis, and what multicollinearity does to your interpretation · Lab	§15.7–15.8	Final Project report due
29	Thu, Dec 10	Final Project presentations — round 1, plus course wrap-up		MindTap Ch 15
–	Dec 14–17	FINAL PROJECT PRESENTATIONS — round 2; day and time per the Registrar's final exam schedule		

What we skip, and why. Eleven of fifteen chapters fit in one semester, plus one section of a twelfth. **Ch 4, Introduction to Probability**, **Ch 5, Discrete Probability Distributions**, and **Ch 6, Continuous Probability Distributions** are omitted, along with **Ch 11, Inference About Population Variances** and the block and factorial designs in §13.4–13.5. None of these appear on either exam, and no probability rules, counting rules, or Bayes' theorem are assessed in this course.

The one exception is §6.2, **the normal distribution**, kept as a single session on Sep 22. Everything from the Central Limit Theorem onward is built on the normal curve, so that section is a prerequisite rather than a topic. What the trade buys is room: three sessions for hypothesis testing, two each for chi-square and ANOVA, and four for regression — the ideas this course is actually assessed on.

Important Dates — Fall 2026

Wed, Sep 9	Last day to add via instructor; last day to self-drop
Wed, Sep 16	Last day to drop via advisor without a "W"
Oct 27 – Nov 17	Priority registration for Spring 2027
Wed, Nov 18	Last day to drop with a "W" via advisor

Nov 23–27	Fall Break — no classes
Dec 14–17	Final Week

Labor Day falls on a Monday and Veterans Day on a Wednesday in 2026, so **this section loses no meetings to holidays** — only Fall Break.

By the Nov 18 withdrawal deadline you will have back the midterm, both projects, and most of your MindTap work — about 610 of 1000 points. That is deliberate: decide with data, not with a feeling. If you are worried about your standing, see me the week of Nov 16.

Core Perspective: Inquiry — Social Sciences (IS)

The IS perspective emphasizes the methods social scientists use to systematically study human behavior. In this course that means **estimation, hypothesis testing, and regression**. The course meets the University IS Core requirement and develops two Core Qualities:

Thinker and Problem Solver. You will reason from evidence gathered through quantitative methods, and apply information literacy — finding, evaluating, using, and creating information. You will define problems, find evidence, conduct analysis, and identify solutions. *Assessed in the interpretation sections of Projects 1 and 2 and the Final Project report.*

Effective Communicator. You will use written and visual communication to create meaning and foster understanding, demonstrating facility in analyzing and interpreting quantitative information. Given large datasets, you will develop approaches to asking and answering questions, then use written and visual communication to inform sound business decisions. *Assessed in the Final Project written report and presentation.*

Policies

Attendance. Expected, not separately graded, and close to determinative — this course moves in chapter order and each chapter leans on the one before it. If you miss a session, get the material from a classmate and from Canvas *before* office hours, so we can spend that time on what you actually do not understand.

Devices. Laptops required on lab days, welcome otherwise. Phones away during exams.

Participation. Ask questions when you are lost, including the ones you think are too basic. In a statistics classroom the "obvious" question is usually the one half the room needs answered.

Missed work. Projects: late work is not accepted. MindTap: no make-ups; the dropped quiz and assignment are the policy. Midterm: documented, university-sanctioned conflicts arranged a week in advance. Presentations: your team presents on its assigned day. For an emergency that makes advance notice impossible, contact me as soon as you are able.

University activities and religious observance. If you will miss class for athletics, a conference, field work, military service, or a religious observance, give me documentation early in the semester — or within the first two weeks for a religious conflict — and we will plan around it without penalty.

Incomplete grades. An "I" is available only when a documented emergency prevents you from finishing a small remaining portion of the course and your work to that point is passing. It is not an alternative to withdrawing.

Academic integrity. All work is individual except the Final Project. Students must abide by the Student Conduct Code and Academic Integrity Guidelines at montana.edu/policy/student_conduct. Paraphrasing or quoting another's work without citation is misconduct, as is relying heavily on unacknowledged source material. Do not share your files or accept files from others; work is checked for plagiarism. Sharing files or cheating on a quiz results in a zero on that work or in the class and a report to the Dean. Within a Final Project team, collaboration is expected and shared files are the point; across teams the rule is unchanged. Copying a workbook, fabricating or altering data, and misreporting results are all violations — the last one especially, because it is the specific dishonesty this discipline exists to prevent.

Generative AI. AI tools are **permitted for learning and not permitted for producing submitted work**.

- *Permitted:* explaining a concept, decoding an Excel error, asking what a *p*-value means, debugging a formula you wrote.
- *Not permitted:* generating your analysis, writing your interpretation, drafting your report, or answering a quiz or exam question.

Every project requires a short **disclosure** of what you used AI for, and you must be able to **explain, in your own words and on request, every number and sentence you submit**. I do ask. Undisclosed AI-generated work is handled as academic misconduct. The reason is not suspicion: the skill this course certifies is judging whether output is trustworthy, and that is exactly the skill you lose by letting a model produce output unexamined.

Disability accommodations. If you have a documented disability for which you may request accommodations, tell me as soon as possible and contact **MSU Disability Services** (Romney Hall · 406-994-2824 · drv@montana.edu · montana.edu/disabilityservices). Accommodations cannot be applied retroactively to submitted work. If you suspect you need accommodations but have never been assessed, Disability Services can advise you.

Mental health and wellness. Your mental health matters as much as your physical health, and it is OK to experience difficulty. Please use these, and tell me if something is affecting your work — no details needed.

Counseling & Psychological Services	montana.edu/counseling · 406-994-4531
Office of Health Advancement	montana.edu/oha
Suicide Prevention	montana.edu/suicide-prevention
Medical Services	montana.edu/health/medical.html
Mental Health Screening	screening.mentalhealthscreening.org/montanastateuniv

Title IX. MSU prohibits discrimination and harassment, including sexual misconduct. As an instructor I am a **mandatory reporter** and must report disclosures of sexual misconduct to the Title IX Coordinator. For a

confidential conversation instead, contact the **VOICE Center** (406-994-7069, 24/7) or Counseling & Psychological Services.

Basic needs. If affording food or securing housing is a problem, MSU has resources including the student food pantry, and the Dean of Students office can help you navigate them. Tell me if it is affecting your coursework and I will help, confidentially.

Name and pronouns. Tell me the name and pronouns you want used in class, whatever appears on my roster.

Veterans and service members. Let me know so I can support you around drill weekends, deployments, and VA appointments.

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Communication. Email is the official channel. Use your MSU address, put **BMGT 240IS** in the subject line, write in complete sentences with a salutation and a signature block (name, course, section), and be concise — I aim to respond within one business day on weekdays. This is a business course and email is a business skill. Please do not ask for what the syllabus already answers, and do not email about a specific grade — that conversation happens in office hours.

Grade concerns. Email me within **one week** of receiving a grade with the specific item and the specific reason. Later disputes go through the department's grade appeal process. Grades are protected under FERPA and posted only in Canvas.

Changes. I may adjust the schedule if the class needs more time on something; changes are announced in class and posted on Canvas. Grade weights and exam dates will not change except in an emergency.

How to Do Well

Statistics is learned by doing it, not by watching it. **Read ahead** — twenty minutes with the chapter before class changes what the lecture is for.

Treat surprise as information. When a result contradicts what you expected, that gap is the most valuable thing in the room, so bring it to class.

Ask what a number means, not just how to compute it. By December you will produce a regression output in forty seconds; the whole game is what you say next.

I am here to help you learn, but you are ultimately responsible for your performance.

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